AIML Phase 1 — Project Report

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Repository: Open GitHub repository | Git tag: phase-1

Executive summary

This phase covered data quality auditing, feature engineering, baseline attrition modelling, risk evaluation, and attendance-anomaly validation. The work includes reproducible scripts, automated tests, CSV evidence, charts, and written interpretations. Results are experimental and require independent validation before operational use.

Task completion and evidence

Task	Work completed	Evidence
1. Data audit	Reviewed data quality, duplicate employee/date pairs, invalid values, target distribution, and planted issues.	reports/data-audit.md; reports/data-audit-overview.png
2. Feature engineering	Built preprocessing and monthly aggregation; tested model-ready features and leakage exclusions.	src/features/; TASK2_EVIDENCE.md; tests/
3. Baseline modelling	Compared Dummy, balanced Logistic Regression, and depth-4 Decision Tree with stratified five-fold validation.	reports/baseline-report.md; baseline_metrics.csv; benchmark comparison
4. Attrition evaluation	Generated out-of-fold scores, classification metrics, calibration/lift charts, and risk bands.	reports/attrition_evaluation.md; metrics; charts
5. Attendance anomalies	Evaluated Isolation Forest and attendance rules against labelled anomaly examples.	reports/attendance-validation-report.md; metrics and type chart

Task 1 — Data quality

The attrition dataset contains 2,389 rows and 42 columns, with 2,383 unique employee IDs and six duplicate employee/date pairs. There are 349 recorded leavers (14.61%); 213 invalid WFH values were flagged for review. The audit documented questionable values rather than silently treating them as correct.

Task 2 — Feature pipeline

The implementation separates feature preparation, preprocessing, and monthly attendance/payroll aggregation. Numeric and categorical preprocessing is used, and target/leakage columns are excluded. The feature and pipeline tests passed.

Task 3 — Baseline modelling

Model	ROC-AUC	PR-AUC
Dummy (always stays)	0.500	0.146
Logistic Regression	0.707	0.318
Decision Tree (depth 4)	0.625	0.226

Logistic Regression achieved ROC-AUC 0.7068 and PR-AUC 0.3176 in the reported cross-validation results. The Dummy model is the no-skill reference; the depth-4 Decision Tree achieved ROC-AUC 0.6247. Detailed benchmark comparison is available in the repository.

Task 4 — Attrition metrics and risk bands

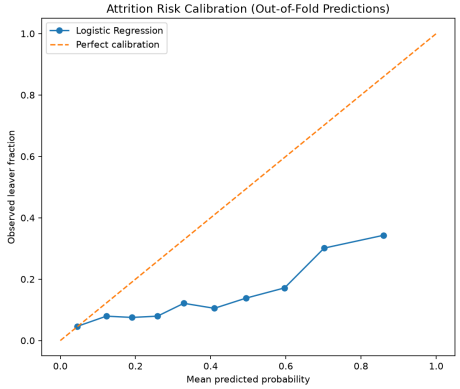
Metric	Result	Interpretation
Accuracy	0.686	Overall classification accuracy
Precision	0.256	Share of predicted leavers who left
Recall	0.599	Share of actual leavers identified
F1	0.358	Balance of precision and recall
ROC-AUC	0.696	Ranking discrimination
PR-AUC	0.291	Precision-recall performance
Brier score	0.208	Probability error; lower is better

These are five-fold out-of-fold probabilities. At threshold 0.5, recall was 0.5989 and precision was 0.2555. PR-AUC is useful because leavers are the minority class; accuracy alone can hide weak minority-class detection.

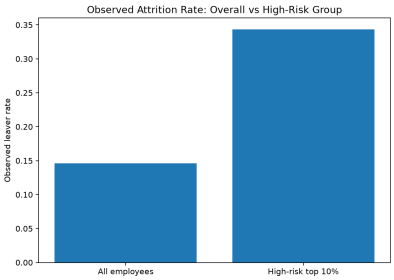
Risk-band analysis

Risk band	Employees	Leavers	Observed leaver rate	Share of leavers captured
High	239	82	34.3%	23.5%
Medium	478	113	23.6%	32.4%
Low	1672	154	9.2%	44.1%

The High-risk band contains 239 employees; 82 were actual leavers (34.31%), about 2.35 times the overall historical leaver rate of 14.61%. However, 154 actual leavers were in the Low band, so risk bands are not guarantees.



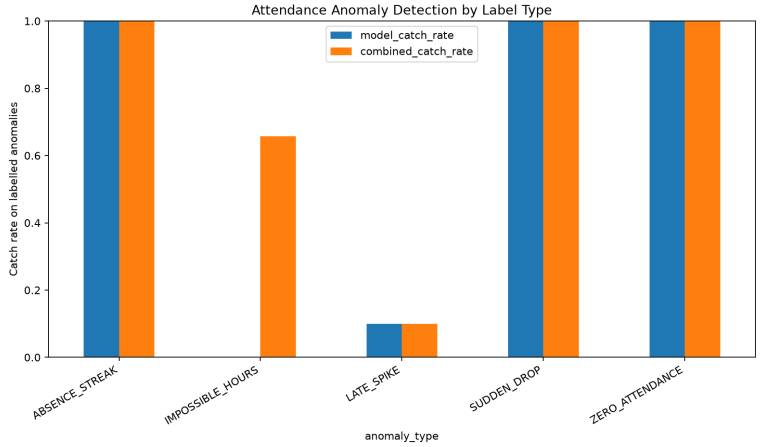
Calibration: points nearer the diagonal indicate closer agreement between predicted probabilities and observed rates. Interpret cautiously because of finite sample sizes.



Task 5 — Attendance anomaly validation

Method	Caught	Missed	Catch rate
Isolation Forest only	88	62	58.7%
Isolation Forest + rules	111	39	74.0%

On 150 labelled anomaly examples, Isolation Forest caught 88 (58.67%) and missed 62. Adding attendance rules caught 111 (74.00%) and missed 39. The rules added 23 detections. Late-arrival spikes remain a weakness: only 3 of 30 were caught by either method. The type-level breakdown is in the CSV evidence.



Testing and reproducibility

The automated test suite completed with 5 passed tests. It covers feature engineering, model-ready output, target/leakage exclusion, pipeline save/load, and the error raised when transforming before fit. Scripts and CSV/Markdown evidence support re-running the evaluations.

Limitations and responsible use

The attendance validation sheet contains labelled anomaly examples but no matched set of confirmed normal records, so catch rate can be measured but precision and false-alarm rate cannot. Workbook baseline precision/recall values are references, not reproduced results. Attrition scores are estimates, not proof of future behaviour. Use them only for voluntary, human-reviewed support discussions, never as stand-alone employment decisions. Further independent validation and fairness checks are needed.

Learning and next steps

This phase strengthened skills in data auditing, preprocessing, cross-validation, probability-based evaluation, risk ranking, anomaly detection, testing, and evidence documentation. Next, I plan to improve late-arrival anomaly rules, obtain labelled normal examples, assess calibration and fairness on future data, and package the pipeline for repeatable deployment.